MULTIMEDIA UNIVERSITY OF KENYA

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

UNIVERSITY EXAMINATIONS 2020/2021

SECOND YEAR FIRST SEMESTER SUPPLEMENTARY/SPECIAL EXAMINATIONS

FOR THE DEGREE OF

BACHELOR OF SCIENCE IN COMPUTER SCIENCE/ BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

CCS 2213/ BIT 2110: OPERATING SYSTEMS II

DATE: 21ST JUNE 2022 TIME: 2 HOURS

INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED.

ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS.

QUESTION ONE (THIRTY MARKS)

a) In concurrency systems, several liveness issues arise. Describe the following issues. You can use an algorithm.

i) Livelocks (3 marks)

ii) Deadlocks (3 marks)

b) When the average queue is only one, the disk scheduling algorithms reduces to FCFS scheduling. *Explain why this assertion is true.* (2marks)

c) Using the layered approach explain the organization of a computer system (4 marks)

d) Compare and contrast cluster and symmetrical multiprocessing. **(8 marks)**

e) Using suitable example, define remote procedure call. **(2 marks)**

f) Draw a process state transition diagram using the five states and explain the interpretation of each process **(5 marks)**

g) Describe how memory and process management are accomplished by the operating system? **(3 marks)**

QUESTION TWO (TWENTY MARKS)

a) In context of multithreading systems, explain the importance of the following and briefly describe how it can be carried out in Java. **(12 marks)**

i) Thread Synchronization

ii) Setting thread priority

iii) Pausing thread

iv) Blocking and unblocking thread using Inter-thread communication

b) **Referring to the figure below answer the following questions:**

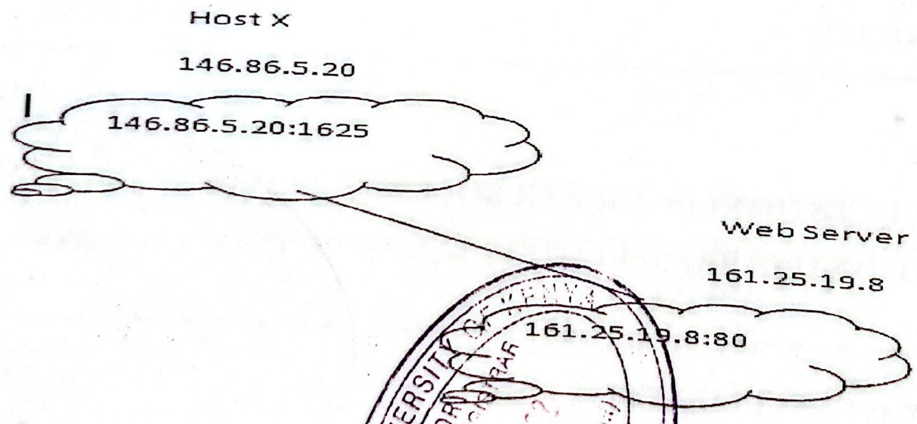

Fig 1.

i) **Briefly describe how interaction takes place between the two systems** **(3 marks)**

ii) What do the 1625 and 80 refer to and why are the necessary in inter-process communication **(3 marks)**

iii) Explain the use and importance of 146.86.20.5 and 161.25.19.8 in the above system **(2 marks)**

QUESTION THREE (TWENTY MARKS)

a) Briefly outline the main purpose of IPC **(2 marks)**

b) The figure below shows two processes T1 and T2 executing in the CPU.

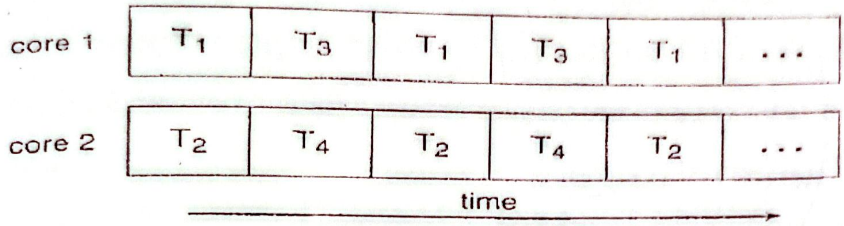

Fig 2.

i) Describe the type of system exhibited in the figure above. **(3 marks)**

ii) Give two advantages of the system stated above. **(2 marks)**

c) Describe the Process control block and the various pieces of information that it contains in regard to process management. **(5 marks)**

d) Describe the Execution of a Remote Procedure Call **(8 marks)**

QUESTION FOUR (TWENTY MARKS)

(a) Differentiate between connection oriented communication and connectionless communication. **(4 marks)**

(b) Explain two roles a timer plays during the execution of a process. **(4 marks)**

(c) Briefly explain what IO subsystem is and describe its role in a computer system **(6 marks)**

(d) Write a program in C/C++ which maliciously access file named studentfee.txt and clear all existing content and replace it with your nickname. Explain working of your program. **(6 marks)**

QUESTION FIVE (TWENTY MARKS)

(a) With aid of a diagram differentiate between local and remote procedure call. **(4 marks)**

(b) Using a suitable diagram and the von Neumann architecture of computers as a reference; describe how the execution of system call takes place in a dual mode operation. **(6 marks)**

(c) Below is a set of processes available for execution in a multi-programmed environment:

Process	Burst Time (Relative Units)	Arrival time at Ready State
A	10	0
B	6	1
C	4	3
D	1	4

a) Show that using Shortest Time Next (SRTN) scheduling algorithm will result in better values for waiting time and turnaround time for each of the process compared to First Come First Served (FCFS). **(6 marks)**

b) Discuss any two facilities to handle system calls errors. **(4 marks)**

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FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

UNIVERSITY EXAMINATIONS 2019/2020

SECOND YEAR FIRST SEMESTER EXAMINATION FOR THE DEGREE OF

BACHELOR OF SCIENCE IN SOFTWARE ENGINEERING

CCS 2213: OPERATING SYSTEMS II

DATE: 13TH DECEMBER 2019 **TIME: 2 HOURS**

INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED.

ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS

QUESTION ONE (THIRTY MARKS)

a) Explain two general roles of an operating system, and elaborate why these roles are important. [4 Marks]

b) Explain the need for distributed operating systems and why they have become more important today [3 Marks]

c) Explain processes in operating systems [2 Marks]

d) Explain threads in operating systems [2 Marks]

e) Explain is a race condition using an example. [3 Marks]

f) Multi-programming (or multi-tasking) enables more than a single process to apparently execute simultaneously. How is this achieved on a single processor machine[2 Marks]

g) State an advantages and disadvantage of user-level threads. [2 Marks]

h) Explain a critical region and howit relates to controlling access to shared resources [2 Marks]

i) What is *deadlock?* [2 Marks]

j) Assuming the operating system detects the system is deadlocked, what can the operating system do to recover from deadlock? [2 Marks]

k) Explain the importance of disabling interrupts during interprocess communication [3 Marks]

l) Explain the message passing method in interprocess communication [3 Marks]

QUESTION TWO (TWENTY MARKS)

a) Write some basic C code that creates a process. It should show whether it's a parent, child or no process created. [6 Marks]

b) Explain four principal events that cause processes to be created [4 Marks]

c) Explain four conditions for process termination [4 Marks]

d) Explain two conditions under which a process may not be created [4 Marks]

e) A *semaphore* is a blocking synchronization primitive. Explain how they work [2 Marks]

QUESTION THREE (TWENTY MARKS)

a) Write some basic C code that creates a thread. [6 Marks]

b) Using an application example of your choice explain how threads help in its task execution [4 Marks]

c) Under what two conditions can a thread **not** be created [2 Marks]

d) Explain three arguments of having threads [6 Marks]

e) Explain two states that a thread can be in [2 Marks]

QUESTION FOUR (TWENTY MARKS)

a) Write some basic C code to demonstrate RPC (remote procedure call) [8 Marks]

b) Netflix is the world largest online movie provider. Explain how it uses replication to achieve a 24/7 uptime experience [5 Marks]

c) Explain security management in distributed systems [4 Marks]

d) Explain the concept of shared memory [3 Marks]

QUESTION FIVE (TWENTY MARKS)

a) Explain three issues that that developers in an organization must do to ensure a well structured InterProcess communication [6 Marks]

b) Explain four conditions required in order to prevent race conditions. [2 Marks]

c) Write some basic C code to demonstrate a semaphore [6 Marks]

d) Explain the concept of distributed file systems [6 Marks]

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FACULTY OF COMPUTING & INFORMATION TECHNOLOGY
UNIVERSITY EXAMINATIONS 2020/2021
SECOND YEAR FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN COMPUTER SCIENCE & SOFTWARE ENGINEERING

ICS 2213: OPERATING SYSTEMS II

DATE: 9TH DECEMBER, 2021 **TIME: 2 HOURS**

INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED

ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS

QUESTION ONE (THIRTY MARKS)

a) Describe the difference between network operating system and distributed operating system **(4 Marks)**

b) Describe four (4) advantages of distributed systems. **(4 Marks)**

c) Define the following terms:
   i. Middleware **(2 Marks)**
   ii. Remote procedure calls. **(2 Marks)**

d) For each case below, state two types of related operating systems with their respective current versions: **(4 Marks)**
   i. Mobile devices
   ii. Servers

e) Describe the working of Remote Procedure Call (RPC) model and the problems associated with the model. **(4 Marks)**

f) Define the following terms as used in operating systems: **(6 Marks)**
   i. On-chip memory
   ii. Command Line Interpreter (CLI)
   iii. I/O Controller

g) Differentiate the following: **(4 marks)**
   i. A program and a process
   ii. Pages and segments

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QUESTION TWO (TWENTY MARKS)

a) Differentiate asynchronous and synchronous models of distributed systems (4 marks)
b) Describe five (5) kinds of transparency in distributed systems. (5 marks)
c) Define the following terms: (6 marks)
   i. Context switching
   ii. Virtual machine
   iii. Interrupt
d) Explain five (5) reasons why distributed systems are becoming popular in modern businesses.
(5 Marks)

QUESTION THREE (TWENTY MARKS)

a) What distinguishes client/server computing from any other form of distributed data processing?
(4 Marks)
b) Explain the differences between the following networking approaches: (2 Marks)
   i. Thin clients
   ii. Fat clients
c) Explain at least four (4) differences between MS Windows and UNIX File Systems?
(8 Marks)
d) Describe two (2) main issues encountered in the design of distributed systems. (4 Marks)
e) Explain exhaustively the scheme of cache validation in OS.
(2 Marks)

QUESTION FOUR (TWENTY MARKS)

a) Explain the difference between a file and a database.
(2 Marks)
b) Briefly explain four (4) file organization mechanisms.
(4 Marks)
c) Bandwidth and Latency are network performance parameters. Briefly explain their meanings.
(4 Marks)
d) Describe how file access control is handled in UNIX operating systems. (5 Marks)
e) Explain the relationship between a path name and a working directory? List three operations performed on a directory.
(5 Marks)

QUESTION FIVE (TWENTY MARKS)

a) Briefly define the term deadlock (2 marks)
b) Explain the necessary and sufficient conditions to produce a deadlock? (8 marks)
c) Explain briefly the method of deadlock avoidance (5 marks)
d) Describe four (4) ways in which systems might provide for users to protect their files against other users.
(5 marks)

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MULTIMEDIA UNIVERSITY OF KENYA

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

UNIVERSITY EXAMINATIONS 2019/2020

SECOND YEAR SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF MATHEMATICS & COMPUTER SCIENCE

BIT 2110 /CCS 2213: OPERATING SYSTEMS II

DATE: 13TH DECEMBER 2019 TIME: 2 HOURS

INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED

ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS

QUESTION ONE [THIRTY MARKS]

a) Explain why distributed systems are gaining popularity in the modern computing world. [4 Marks]

b) Distinguish between distributed OS and a network OS. Outline the outstanding characteristics which distinguish them. [4 Marks]

c) Web services use an XML based protocol (SOAP) to send requests and responses. What are the advantages and disadvantages of using XML for remote calls? [4 Marks]

d) Describe the following types of systems

   i. Grid computing
   ii. Utility Computing
   iii. Service oriented Architecture
   iv. Cloud services [8 Marks]

e) Differentiate with brief-explanations the following location policies under resource management

   i. Threshold and shortest methods
   ii. Bidding and paring methods [6 marks]

f) Describe the concept of group communication outlining the main schemes used to implement it. [4 Marks]

QUESTION TWO (TWENTY MARKS)

a) Discuss the main issues which must to be considered when designing a good IPC protocol for a message passing system. [4 Marks]

b) Define what a transaction

c) Describe what a transaction in a distributed system is and the ACID properties it ought satisfy. [6 Marks]

d) Explain what interleaving of two transactions is. (2 Marks).

e) With the aid of a suitable diagram, discuss the following DCS models
   i. Workstation
   ii. Processor pool
   iii. Hybrid
   iv. Workstation –server [8 marks]

QUESTION THREE (TWENTY MARKS)

a) Describe how reliability in group communication is assessed. [2 Marks]
b) Describe the term file replication and its main advantages in advanced operating systems [8 Marks]
c) Discuss the main approaches under resource management to schedule processes to processors. [10 marks]

QUESTION FOUR (TWENTY MARKS)

a) Describe the term Remote Procedure Call and with a the aid of a suitable diagram, describe how RPCs are facilitated in advanced operating systems. [10 Marks]
b) Describe the following types of transparency as implemented in distributed systems
   i. Replication
   ii. Failure
   iii. Concurrency
   iv. Scalability

[10 Marks]

QUESTION FIVE (TWENTY MARKS)

a) Describe the design principles considered when ensuring that distributed systems are build for optimum performance. [5 Marks]

b) Describe the functions of the three lower layers of the OSI protocol used in distributed systems. [6 Marks]
c) Explain three main buffering techniques employed when improving communication be advanced operating systems . [3 Marl

d) Outline what is meant by the following ordering of message techniques in group communication

i. Absolute

ii. Consistent

iii. Causal **[6 marks]**